

5	A particle moving with kinetic energy K undergoes a head-on perfectly inelastic collision with an identical particle that is initially at rest. What is the total kinetic energy of both particles, in terms of K, after the collision?			
	A 0.25 K	B 0.5 K	C K	D $2K$
L2	Answer: B Let m be the mass of each particle and u be the initial speed of the first particle. $K = \frac{1}{2} m u^2$ In a perfectly inelastic collision, the two particles stick together and move with a common velocity V after the collision. By principle of conservation of momentum, $mu + 0 = (m + m) V$ $mu = (2m) V$ $V = \frac{1}{2} u$ Thus total KE of both particles after the collision $= \frac{1}{2} (2m) V^2$ $= m (\frac{1}{2} u)^2$ $= \frac{1}{2} (\frac{1}{2} m u^2) = \frac{1}{2} K = \mathbf{0.5 K}$			

6 One end of a spring is fixed to a support. A mass is attached to the other end of the spring as